classification

types of classification

1- Binary

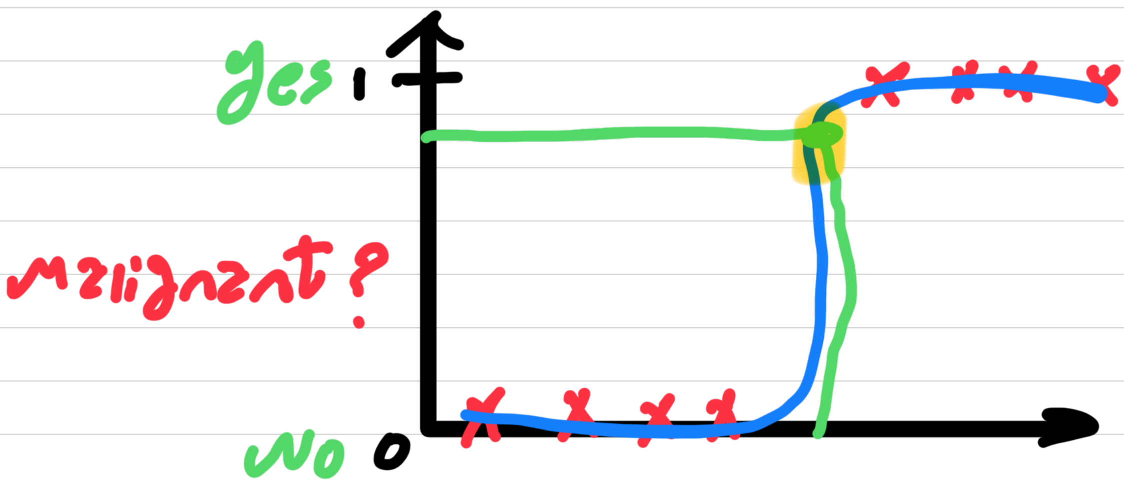
Is the email spam?

Is tumor malignant?

negative class
0 - No - false

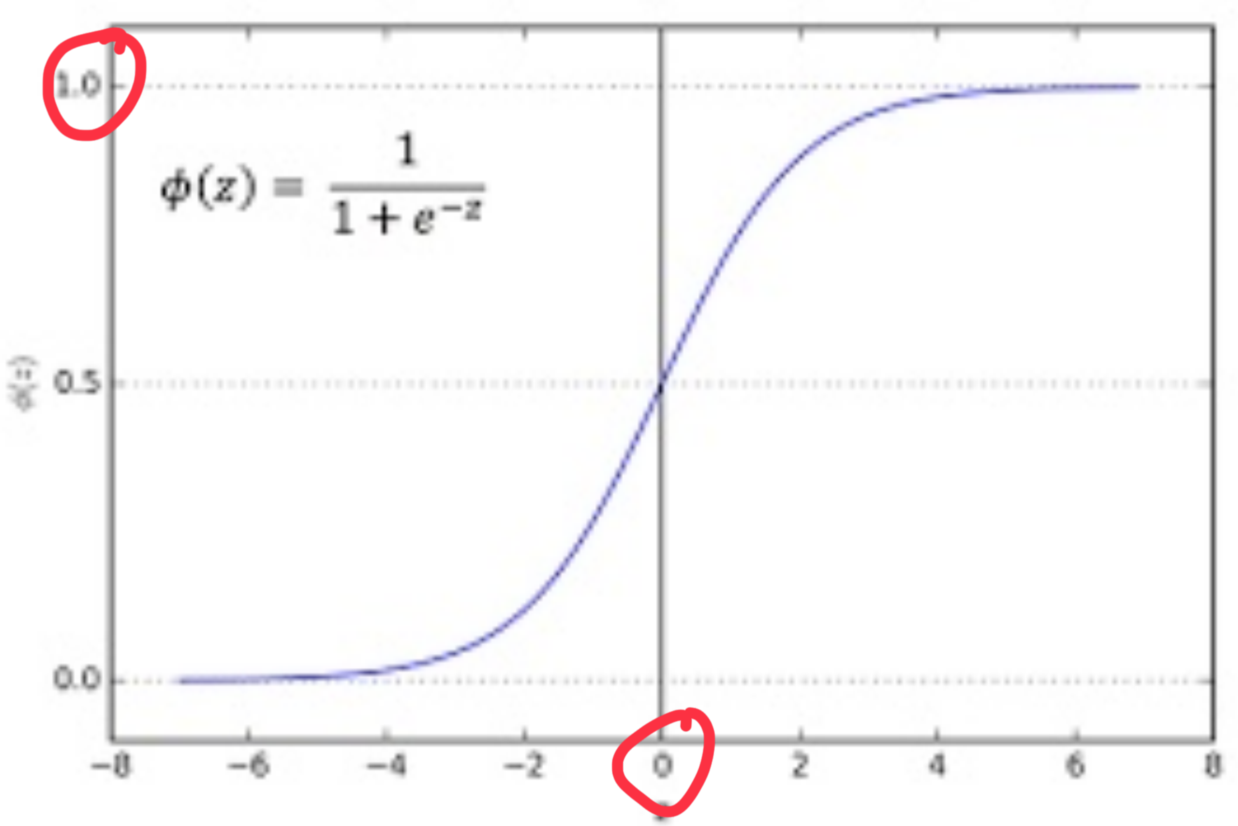
1 - yes - True
Positive class

Logistic Regression



want output between 0,1

sigmoid function



- output between 0,1

$$g(z) = \frac{1}{1+e^{-z}} \quad , \quad 0 < g(z) < 1$$

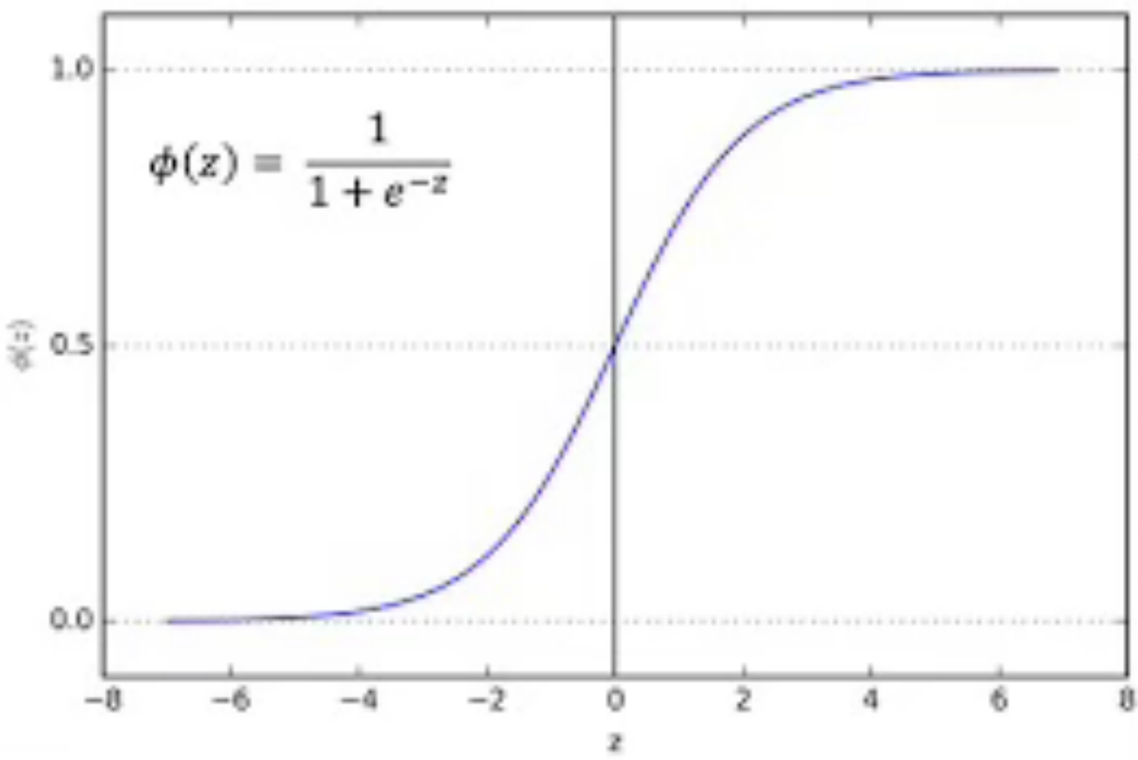
- If z is a large positive num

$g(z) \rightarrow$ is near zero

$z = -100$

then

$e^{100} \rightarrow \frac{1}{1+e^{100}} \rightarrow$ Near zero



$$f_{\vec{w},b}(\vec{x}) = \vec{w} \cdot \vec{x} + b$$

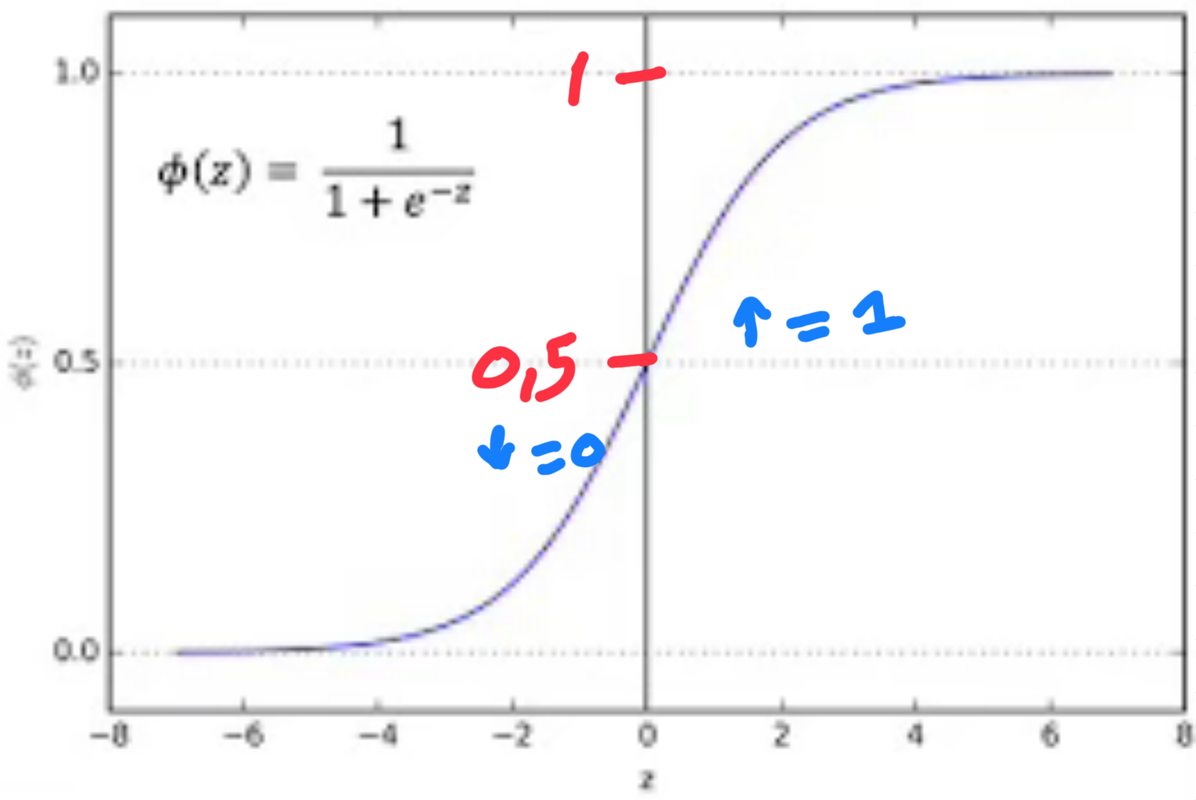
$$z = \vec{w} \cdot x + b$$

$$g(z) = \frac{1}{1+e^{-z}} \qquad 0 < g(z) < 1$$

this gives the logistic R model

$$f_{\vec{w},b}(\vec{x}) = g(\underbrace{\vec{w} \cdot \vec{x} + b}_{z}) = \frac{1}{1+e^{-(w \cdot x + b)}}$$

Decision Boundary



When :

$$\left.\begin{array}{l} g(z) \geq 0{,}5 \\ z \geq 0 \\ \vec{w}\cdot\vec{x}+b \geq 0 \end{array}\right\} \rightarrow y = 1$$

$$\left.\begin{array}{l} z < 0 \\ w\cdot x+b < 0 \end{array}\right] \rightarrow y = 0$$

Linear Decision boundary

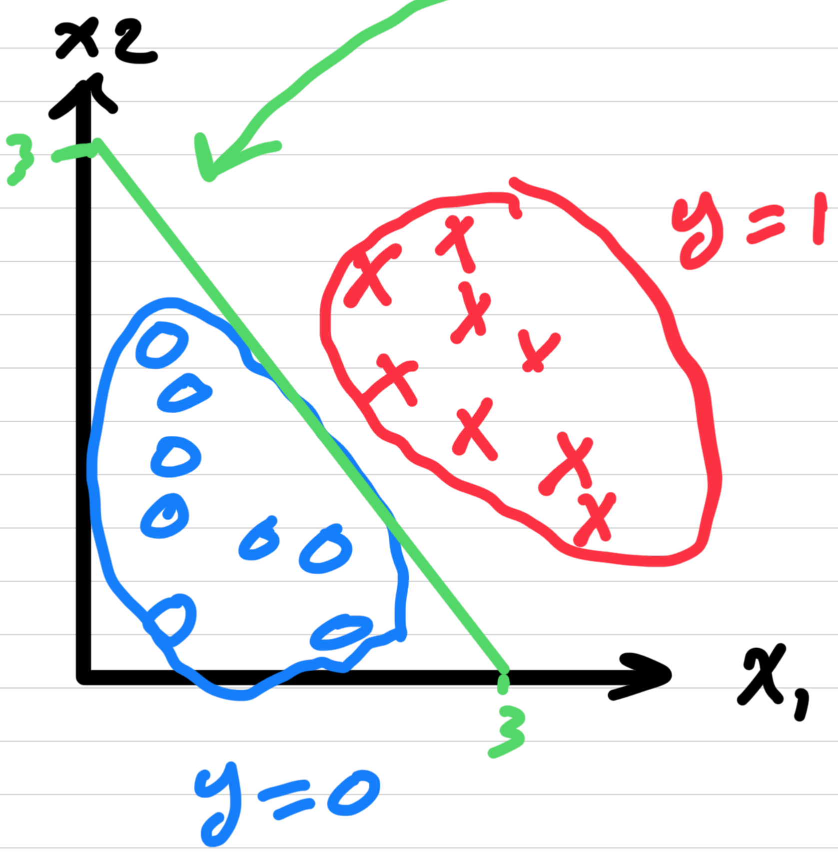
$$f_{w,b} = g(z) = g(\overbrace{w_1 x_1 + w_2 x_2 + b}^{z})$$

$$w_1 = 1 \qquad w_2 = 1 \qquad b = -3$$

Decision boundary: $z = \vec{w} \cdot \vec{x} + b = 0$

$$z = x_1 + x_2 - 3 = 0$$

$$x_1 + x_2 = 3$$



$x_1 + x_2 > 3 \rightarrow y = 1$

$x_1 + x_2 < 3 \rightarrow y = 0$

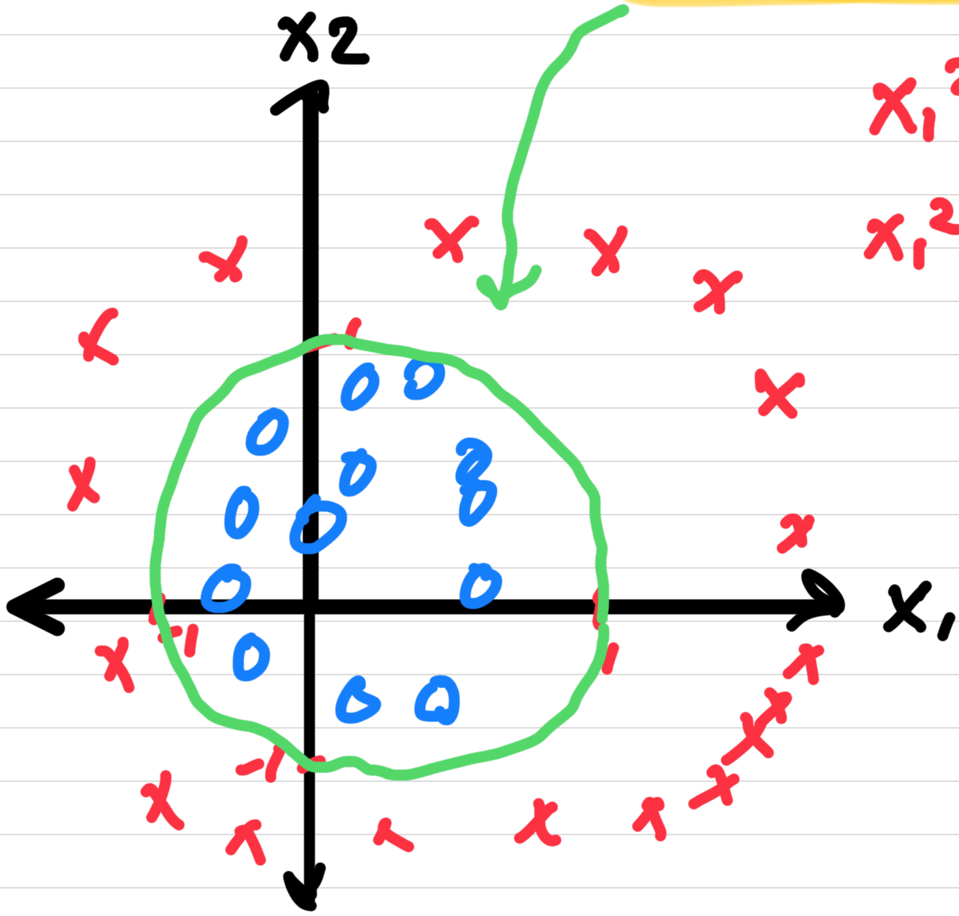
- the green line is the Decision boundary
- If feature X to the Right $\longrightarrow$ $y = 1$

  to the left $\rightarrow$ $y = 0$

$$z = w_1 x_1^2 + w_2 x_2^2 + b$$

$$x_1^2 + x_2^2 - 1 = 0$$

$$x_1^2 + x_2^2 = 1$$

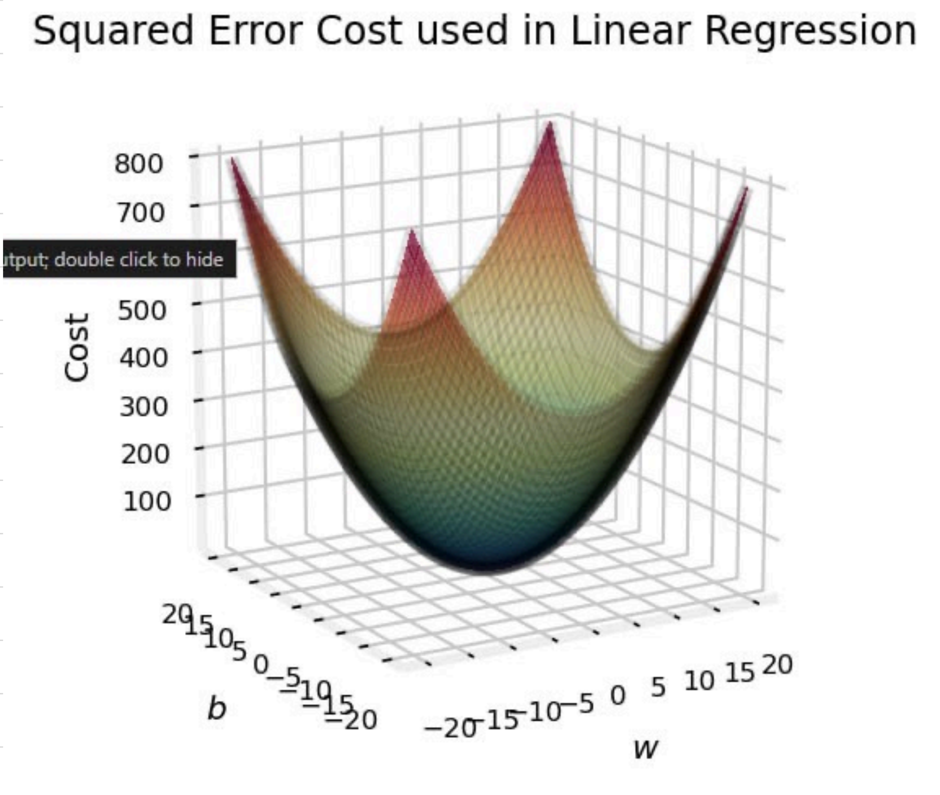


$x_1^2 + x_2^2 > 1 \rightarrow y = 1$

$x_1^2 + x_2^2 < 1 \rightarrow y = 0$

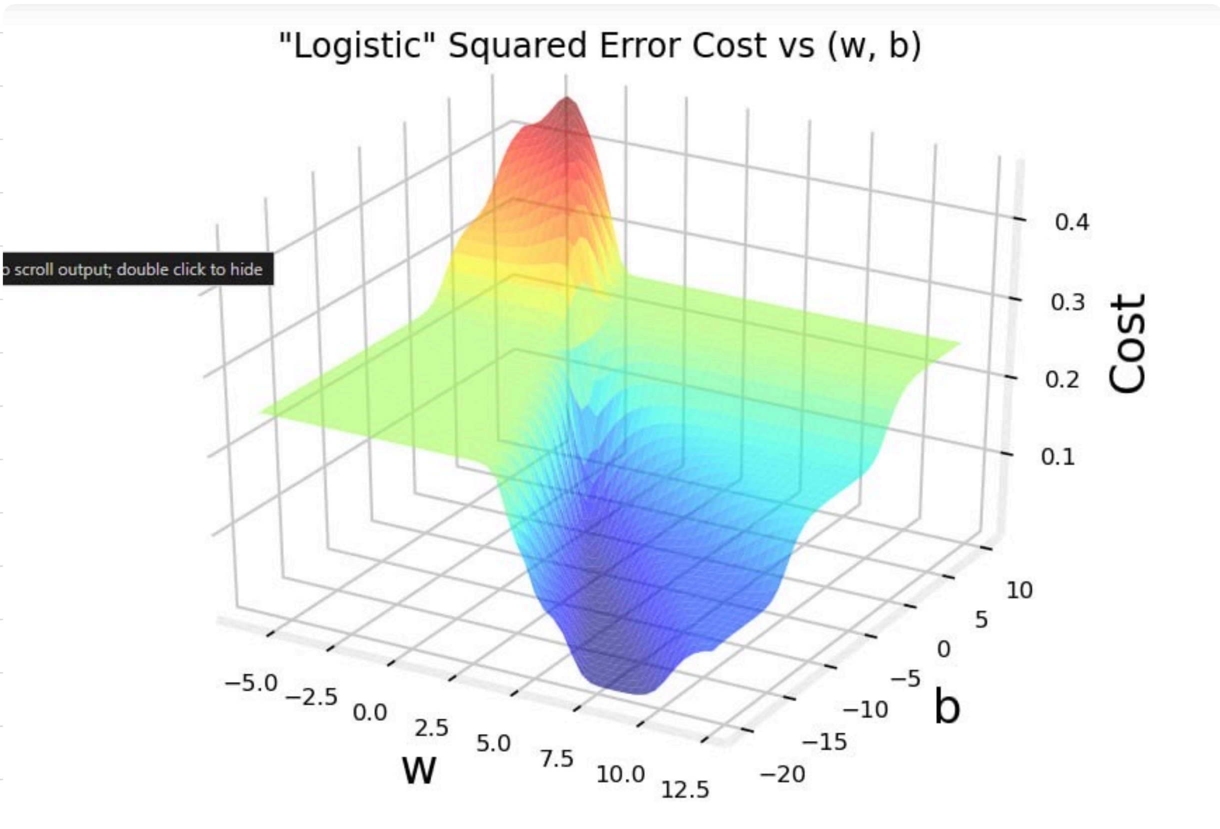
Cost fun for LR

- y : $\in \{0,1\}$ true value class
- $\hat{y}$: predicted $\sigma(z) = \frac{1}{1+e^{-z}}$
- z : $\vec{w} \cdot \vec{x} + b$ our linear model
- Cost : $\frac{1}{2m} \sum_{i=1}^{m} (f_{w,b}(x_i) - y_i)^2$



- this cost fun worked well for Linear R

- If we used it for Logistic Regression



- It will give us this plot which is not smoth as as plot from linear R

Squared Error Cost

cost

$$J(\vec{w},b) = \frac{1}{m}\sum_{i=1}^{m} \underbrace{\frac{1}{2}(\hat{y}^{(i)} - y^{(i)})^2}_{\text{loss}}$$

$\hat{y}^{(i)} = f_{\vec{w},b}(\vec{x}^{(i)})$

average of training set

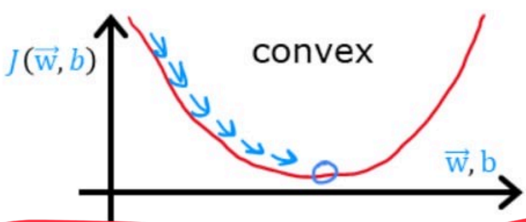
loss

$$L(\hat{y}^{(i)}, y^{(i)}) = \frac{1}{2}(\hat{y}^{(i)} - y^{(i)})^2$$

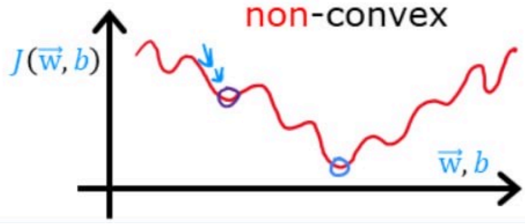
single training example

$$J(\vec{w},b) = \frac{1}{m}\sum_{i=1}^{m} L(\hat{y}^{(i)}, y^{(i)})$$

linear regression $f_{\vec{w},b}(\vec{x}) = \vec{w}\cdot\vec{x} + b$



logistic regression $f_{\vec{w},b}(\vec{x}) = \frac{1}{1+e^{-(\vec{w}\cdot\vec{x}+b)}}$



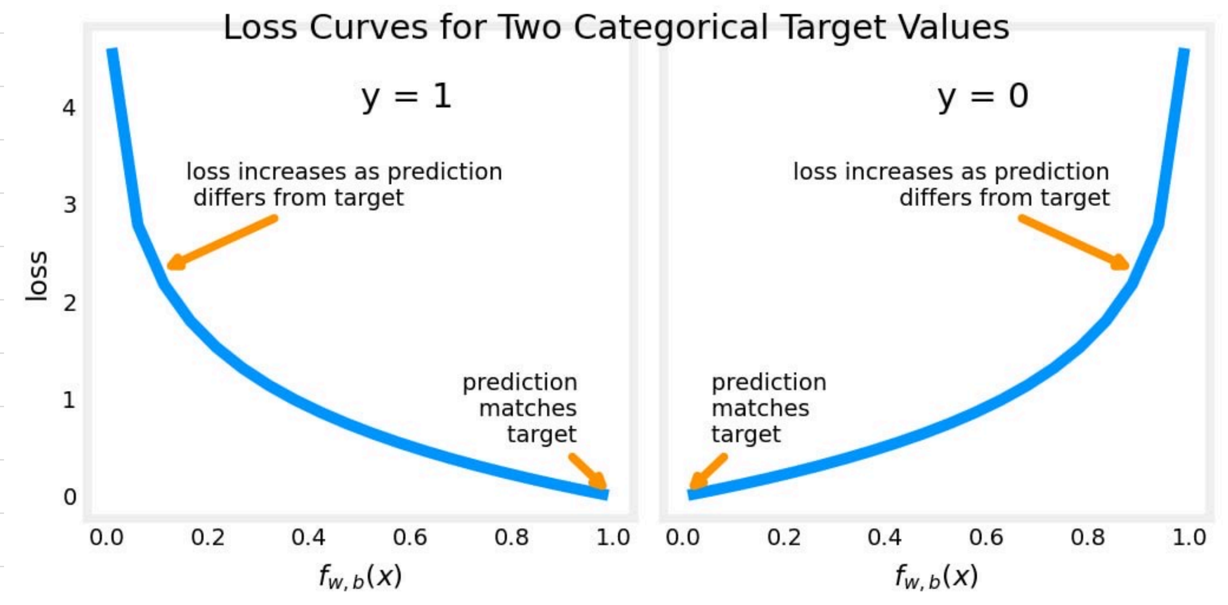
The solution

we use the **log loss**

$$L(f_{w,b}(\vec{x}), y) \begin{cases} -\log(f_{\vec{w},b}(x)) & \text{if } y=1 \\ -\log(1-f_{\vec{w},b}(x)) & y=0 \end{cases}$$

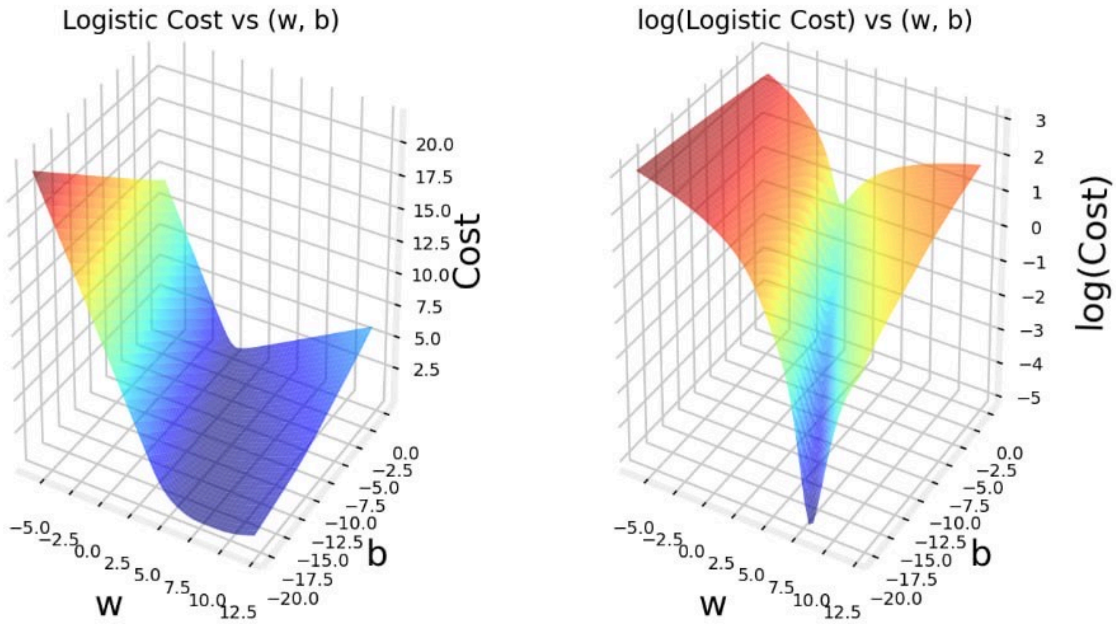
- $f_{w,b}(x_i) \rightarrow \hat{y}$ **predicted**
- $y \rightarrow$ target

- this loss fun uses **two curves** $< \begin{matrix} y=1 \\ y=0 \end{matrix}$



rewritten the cost fun

$$-\underbrace{y_i}_{\text{target}} \log(\underbrace{f_{w,b}(x_i)}_{\text{y-hat}}) - (1-\underbrace{y_i}_{\text{target}}) \log(1-\overbrace{f_{w,b}(x_i)}^{\text{y-hat}})$$


Logistic Cost vs (w, b)
Cost
w
b
log(Logistic Cost) vs (w, b)
log(Cost)
w
b